VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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LAB REPORT on

Big Data Analytics

Submitted by

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CERTIFICATE

This is to certify that the Lab work entitled "Big Data Analytics" carried out by **TEJAS JOSHI (1WA23CS016),** who is a bona fide student of **B. M. S. College of Engineering.** It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2026. The Lab report has been approved as it satisfies the academic requirements in respect of a **Big Data Analytics - (23CS6PCBDA)** work prescribed for the said degree.

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GITHUB LINK: https://github.com/tejasnewbie/BDA-LAB

Course Outcome

CO1	Apply the concepts of NoSQL, Hadoop, Spark for a given task
CO2	Analyze data analytic techniques for a given problem
CO3	Conduct experiments using data analytics mechanisms for a given problem

LAB 1
MongoDB- CRUD Demonstration

Student database

1. Create a database “Student” with the following attributes Rollno, Age, ContactNo, Email-Id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Student");
{ ok: 1 }
```

2. Insert appropriate values (at least 5)

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:1,Age:21,Cont:9876,email:"antara.de9@gmail.com"});
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ec7c840f42b4a05530") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:2,Age:22,Cont:9976,email:"anushka.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05531") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:3,Age:21,Cont:5576,email:"anubhav.de9@gmail.com"});

{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05532") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:4,Age:20,Cont:4476,email:"pani.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82ed7c840f42b4a05533") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a82f47c840f42b4a05534") }
```

3. View the data

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.find()
[
  {
    _id: ObjectId("660a82ec7c840f42b4a05530"),
    RollNo: 1,
    Age: 21,
    Cont: 9876,
    email: 'antara.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05531"),
    RollNo: 2,
    Age: 22,
    Cont: 9976,
    email: 'anushka.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05532"),
    RollNo: 3,
    Age: 21,
    Cont: 5576,
    email: 'anubhav.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82ed7c840f42b4a05533"),
    RollNo: 4,
    Age: 20,
    Cont: 4476,
    email: 'pani.de9@gmail.com'
  },
  {
    _id: ObjectId("660a82f47c840f42b4a05534"),
    RollNo: 10,
    Age: 23,
    Cont: 2276,
    email: 'rekha.de9@gmail.com'
  }
]
```

4. Write a query to update the Email-Id of a student with rollno 10.

```
  },
  {
    _id: ObjectId("660a83337c840f42b4a05535"),
    RollNo: 11,
    Age: 22,
    Name: 'ABC',
    Cont: 2276,
    email: 'rea.de9@gmail.com'
  }
]
```

5. Replace the student name from “ABC” to “FEM” of rollno 11.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.update({RollNo:11,Name:"ABC"},{$set:{Name:"FEM"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

6. Export the created table into local file system

```
C:\Users\BMSCECSE>mongoexport mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB  --collection=Student --out C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:15:56.383+0530   connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:15:56.497+0530   exported 7 records
```

7. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Student.drop();
true
```

8. Import a given csv dataset from local file system into mongodb collection.

```
C:\Users\BMSCECSE>mongoimport  mongodb+srv://antararc:Test1234@cluster0.mfnfeys.mongodb.net/myDB --collection=New_Student  --type json --file C:\Users\BMSCECSE\Downloads\output.json
2023-01-12T15:17:35.523+0530   connected to: mongodb+srv://[**REDACTED**]@cluster0.mfnfeys.mongodb.net/myDB
2023-01-12T15:17:35.640+0530   7 document(s) imported successfully. 0 document(s) failed to import.
```

Customer database

1. Create a collection by name Customers with the following attributes. Cust_id, Acc_Bal, Acc_Type

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.createCollection("Customers");
{ ok: 1 }
```

2. Insert at least 5 values into the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:1,Balance:200, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05536") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:1,Balance:1000, Type:"Z"})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05537") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:100, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b47c840f42b4a05538") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:1000, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a05539") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:500, Type:"C"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553a") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553b") }
}
Atlas atlas-b6pfyk-shard-0 [primary] test>

Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:3,Balance:500, Type:"Z"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b77c840f42b4a0553c") }
}
```

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.insert({cust_id:2,Balance:50, Type:"S"});
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("660a83b57c840f42b4a0553b") }
}
```

3. Write a query to display those records whose total account balance is greater than 1200 of account type ‘Z’ for each customer_id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.aggregate (
... {$match:{Type:"Z"}},
...
... {$group : { _id : "$cust_id",
...
... TotAccBal :{$sum:"$Balance"} } },
... {$match:{TotAccBal:{$gt:1200}}});
```

4. Determine Minimum and Maximum account balance for each customer_id.

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.aggregate (
...
... {$group : { _id : "$cust_id",
...
... minAccBal :{$min:"$Balance"},
... maxAccBal :{$max:"$Balance"} }});
[
  { _id: 2, minAccBal: 50, maxAccBal: 1000 },
  { _id: 1, minAccBal: 200, maxAccBal: 1000 },
  { _id: 3, minAccBal: 500, maxAccBal: 500 }
]
```

5. Drop the table

```
Atlas atlas-b6pfyk-shard-0 [primary] test> db.Customers.drop()
true
```

LAB

2

Perform the following DB operations using Cassandra

Employee Database

1. Create a keyspace by name Employee

```
cqlsh:library> CREATE KEYSPACE Employee WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh:library>
```

2. Create a column family by name Employee-Info with attributes Emp_Id Primary Key,

```
cqlsh:employee>
cqlsh:employee> CREATE TABLE Employee_Info (
           ...     Emp_Id int PRIMARY KEY,
           ...     Emp_Name text,
           ...     Designation text,
           ...     Date_of_Joining date,
           ...     Salary decimal,
           ...     Dept_Name text
           ... );
```

Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name

3. Insert the values into the table in batch

```
cqlsh:employee> BEGIN BATCH
           ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
           ... VALUES (101, 'John Doe', 'Manager', '2023-01-01', 50000, 'HR');
           ... INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name)
           ... VALUES (121, 'Jane Smith', 'Developer', '2023-02-01', 60000, 'IT');
           ... APPLY BATCH;
cqlsh:employee> UPDATE Employee_Info SET Emp_Name = 'Jane Johnson', Dept_Name = 'Engineering' WHERE Emp_Id = 121;
cqlsh:employee> SELECT * FROM Employee_Info;

 emp_id | date_of_joining | dept_name   | designation | emp_name     | salary
--------+-----------------+-------------+-------------+--------------+--------
    121 |      2023-02-01 | Engineering |   Developer | Jane Johnson |  60000
    101 |      2023-01-01 |          HR |     Manager |     John Doe |  50000

(2 rows)
```

4. Update Employee name and Department of Emp-Id 121

5. Sort the details of Employee records based on salary

```
cqlsh:employee> paging off
Disabled Query paging.
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |             {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

6. Alter the schema of the table Employee_Info to add a column Projects which stores a set of Projects done by the corresponding Employee.
7. Update the altered table to add project names.

```
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectA', 'ProjectB'} WHERE Emp_Id = 101 and salary=50000;
cqlsh:employee> UPDATE Employee_Info SET Projects = {'ProjectC'} WHERE Emp_Id = 121 and salary=60000;
cqlsh:employee> select * from Employee_Info;

 emp_id | salary | date_of_joining | dept_name | designation | emp_name   | projects
--------+--------+-----------------+-----------+-------------+------------+--------------------------
    121 |  60000 |      2023-02-01 |        IT |   Developer | Jane Smith |              {'ProjectC'}
    101 |  50000 |      2023-01-01 |        HR |     Manager |   John Doe | {'ProjectA', 'ProjectB'}

(2 rows)
```

8. Create a TTL of 15 seconds to display the values of Employees.

```
cqlsh:employee> INSERT INTO Employee_Info (Emp_Id, Emp_Name, Designation, Date_of_Joining, Salary, Dept_Name) VALUES (102, 'Jane Smit
h', 'Developer', '2022-06-03', 60000, 'IT') USING TTL 15;
cqlsh:employee> select ttl(Emp_Name) from Employee_Info where Emp_id=102;

 ttl(emp_name)
---------------
            14

(1 rows)
```

LAB

3

Perform the following DB operations using Cassandra

1. Create a keyspace by name Library

```
cqlsh> CREATE KEYSPACE Library WITH REPLICATION = { 'class' : 'SimpleStrategy', 'replication_factor' : 1 };
cqlsh> show keyspaces;
Improper show command.
cqlsh> use Library;
cqlsh:library>
```

2. Create a column family by name Library-Info with attributes Stud_Id Primary Key, Counter_value of type Counter, Stud_Name, Book-Name, Book-Id, Date_of_issue

```
cqlsh:library> CREATE TABLE Library_Info (Stud_Id int PRIMARY KEY,Counter_value counter,Stud_Name text,Book_Name text,Book_Id text,Da
te_of_issue timestamp);
InvalidRequest: Error from server: code=2200 [Invalid query] message="Cannot mix counter and non counter columns in the same table"
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id int PRIMARY KEY,
           ...     Stud_Name text,
           ...     Book_Name text,
           ...     Book_Id text,
           ...     Date_of_issue timestamp
           ... );
cqlsh:library> CREATE TABLE Library_Counters (
           ...     Stud_Id int PRIMARY KEY,
           ...     Counter_value counter
           ... );
cqlsh:library>
```

3. Insert the values into the table in batch

```
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) VALUES (112, 'John Doe', 'BDA', 'B001
', '2023-01-01');
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) VALUES (113, 'Jane Smith', 'ML', 'B00
2', '2023-01-02');
           ... APPLY BATCH;
```

4. Display the details of the table created and increase the value of the counter

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue                   | stud_name
---------+---------+-----------+---------------------------------+------------
     113 |    B002 |        ML | 2023-01-02 00:00:00.000000+0000 | Jane Smith
     112 |    B001 |       BDA | 2023-01-01 00:00:00.000000+0000 |   John Doe

(2 rows)
cqlsh:library> SELECT * FROM Library_Counters;

 stud_id | counter_value
---------+---------------
     113 |             1
     112 |             1

(2 rows)
```

5. Write a query to show that a student with id 112 has taken a book “BDA” 2 times.

```
cqlsh:library> UPDATE Library_Counters SET Counter_value = Counter_value + 1 WHERE Stud_Id = 112;
cqlsh:library> SELECT * FROM Library_Counters WHERE Stud_Id = 112;

 stud_id | counter_value
---------+---------------
     112 |             2

(1 rows)
```

6. Export the created column to a csv file

```
cqlsh:library> COPY Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue) TO 'file.csv' WITH HEADER = TRUE;
Using 11 child processes

Starting copy of library.library_info with columns [stud_id, stud_name, book_name, book_id, date_of_issue].
Processed: 2 rows; Rate:      10 rows/s; Avg. rate:       6 rows/s
2 rows exported to 1 files in 0.374 seconds.
cqlsh:library> COPY Library_Counters (Stud_Id, Counter_value) FROM 'library_counters.csv' WITH HEADER = TRUE;
Using 11 child processes
```

7. Import a given csv dataset from local file system into Cassandra column family

```
cqlsh:library> copy library_info(Stud_Id,Stud_Name,Book_Name,Book_Id,Date_of_issue) from 'file.csv' with header=true;
Using 7 child processes

Starting copy of library.library_info with columns [stud_id, stud_name, book_name, book_id, date_of_issue].
Processed: 2 rows; Rate:       2 rows/s; Avg. rate:       4 rows/s
2 rows imported from 1 files in 0.513 seconds (0 skipped).
cqlsh:library> select * from library_info;

 stud_id | book_id | book_name | date_of_issue                   | stud_name
---------+---------+-----------+---------------------------------+-----------
     113 |    B002 |        ML | 2023-01-02 00:00:00.000000+0000 | Jane Smith
     112 |    B001 |       BDA | 2023-01-01 00:00:00.000000+0000 |   John Doe
```

LAB

4

Execution of HDFS Commands for interaction with Hadoop Environment. (Minimum 10 commands to be executed)

1. mkdir
2. ls

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-800-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -mkdir /bda_hadoop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /
Found 1 items
drwxr-xr-x   - hadoop supergroup          0 2024-05-13 14:37 /bda_hadoop
```

3. put

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -put /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /bda_hadoop
Found 1 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /bda_hadoop/file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file.txt
Hello!!!
```

4. copyFromLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyFromLocal /home/hadoop/Desktop/bda_local.txt /bda_hadoop/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

5. get

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -get /bda_hadoop/file.txt /home/hadoop/Desktop/downloaded_file.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -getmerge /bda_hadoop/file.txt /bda_hadoop/file_cp_local.txt /home/hadoop/Desktop/downloaded_file.txt
```

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -getfacl /bda_hadoop/
# file: /bda_hadoop
# owner: hadoop
# group: supergroup
user::rwx
group::r-x
other::r-x
```

6. copyToLocal

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -copyToLocal /bda_hadoop/file.txt /home/hadoop/Desktop
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

7. cat

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hdfs dfs -cat /bda_hadoop/file_cp_local.txt
Hello!!!
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

8. mv

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -mv /bda_hadoop /abc
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -ls /abc
Found 2 items
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:42 /abc/file.txt
-rw-r--r--   1 hadoop supergroup          9 2024-05-13 14:52 /abc/file_cp_local.txt
```

9. cp

```
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$ hadoop fs -cp /hello/ /hadoop_lab
cp: `/hello/': No such file or directory
hadoop@bmscecse-HP-Elite-Tower-800-G9-Desktop-PC:~$
```

LAB

5

Implement WordCount Program on Hadoop framework

Word Count Program:

Mapper code: You have to copy paste this program into the WCMapper Java Class file

```
import java.io.IOException; import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.LongWritable; import org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.MapReduceBase; import
org.apache.hadoop.mapred.Mapper; import
org.apache.hadoop.mapred.OutputCollector; import
org.apache.hadoop.mapred.Reporter;
public class WCMapper extends MapReduceBase implements Mapper<LongWritable, Text,
Text, IntWritable> { public void map(LongWritable key, Text value, OutputCollector<Text,
    IntWritable> output,
Reporter rep) throws IOException { String line =
      value.toString(); for (String word : line.split(" ")) { if
      (word.length() > 0) { output.collect(new Text(word),
      new IntWritable(1)); }
      }
   }
}
```

Reducer Code: You have to copy paste this program into the WCReducer Java Class file

```
import java.io.IOException; import java.util.Iterator; import
org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import
org.apache.hadoop.mapred.MapReduceBase; import
org.apache.hadoop.mapred.OutputCollector; import
org.apache.hadoop.mapred.Reducer; import org.apache.hadoop.mapred.Reporter;
public class WCReducer extends MapReduceBase implements Reducer<Text, IntWritable, Text,
IntWritable> { public void reduce(Text key, Iterator<IntWritable> value, OutputCollector<Text,
    IntWritable>
output, Reporter rep) throws IOException { int
      count = 0;
      while (value.hasNext()) {
         IntWritable i = value.next();
         count += i.get();
      } output.collect(key, new
   IntWritable(count)); }
}
```

Driver Code: You have to copy paste this program into the WCDriver Java Class file.

```
import java.io.IOException; import org.apache.hadoop.conf.Configured; import
org.apache.hadoop.fs.Path; import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import org.apache.hadoop.mapred.FileInputFormat;
```

```
import org.apache.hadoop.mapred.FileOutputFormat; import
org.apache.hadoop.mapred.JobClient; import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool; import org.apache.hadoop.util.ToolRunner;
public class WCDriver extends Configured implements Tool { public int run(String[]
args) throws IOException { if (args.length < 2) {
        System.out.println("Please give valid inputs"); return
        -1;
    }
    JobConf conf = new JobConf(WCDriver.class); conf.setJobName("Word
    Count");
    FileInputFormat.setInputPaths(conf, new Path(args[0]));
    FileOutputFormat.setOutputPath(conf, new
    Path(args[1])); conf.setMapperClass(WCMapper.class);
    conf.setReducerClass(WCReducer.class);
    conf.setMapOutputKeyClass(Text.class);
    conf.setMapOutputValueClass(IntWritable.class);
    conf.setOutputKeyClass(Text.class);
    conf.setOutputValueClass(IntWritable.class);
    JobClient.runJob(conf); return 0;
  } public static void main(String[] args) throws Exception
  {
    int exitCode = ToolRunner.run(new WCDriver(), args);
    System.out.println(exitCode);
  }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input_file.txt
Hello World
Hello Hadoop
This is Hadoop test file
C:\hadoop-3.3.0\sbin>hadoop jar C:\MapReduceClient.jar wordcount /input_dir /output_dir
2021-04-24 15:24:57,242 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-04-24 15:24:57,714 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging
/job_1619256355508_0002
2021-04-24 15:24:58,387 INFO input.FileInputFormat: Total input files to process : 1
2021-04-24 15:24:58,809 INFO mapreduce.JobSubmitter: number of splits:1
2021-04-24 15:24:59,255 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1619256355508_0002
2021-04-24 15:24:59,255 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-04-24 15:24:59,450 INFO conf.Configuration: resource-types.xml not found
2021-04-24 15:24:59,451 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-04-24 15:24:59,533 INFO impl.YarnClientImpl: Submitted application application_1619256355508_0002
2021-04-24 15:24:59,581 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1619256355508_0002/
2021-04-24 15:24:59,582 INFO mapreduce.Job: Running job: job_1619256355508_0002
2021-04-24 15:25:12,857 INFO mapreduce.Job: Job job_1619256355508_0002 running in uber mode : false
2021-04-24 15:25:12,861 INFO mapreduce.Job:  map 0% reduce 0%
2021-04-24 15:25:19,985 INFO mapreduce.Job:  map 100% reduce 0%
2021-04-24 15:25:26,077 INFO mapreduce.Job:  map 100% reduce 100%
2021-04-24 15:25:32,181 INFO mapreduce.Job: Job job_1619256355508_0002 completed successfully
2021-04-24 15:25:32,284 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=85
                FILE: Number of bytes written=530945
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=162
                HDFS: Number of bytes written=51
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
Hadoop  2
Hello   2
This    1
World   1
file    1
is      1
test    1

C:\hadoop-3.3.0\sbin>
```

LAB 6

From the following link extract the weather data
https://github.com/tomwhite/hadoop-Book/tree/master/input/ncdc/all

Create a Map Reduce program to
a) find average temperature for each year from the NCDC data set.

AverageDriver package temp; import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapreduce.Job; import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat; import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat; public
class AverageDriver { public static void main(String[] args) throws
Exception { if (args.length != 2) {
System.err.println("Please Enter the input and output parameters"); System.exit(-1);
}
Job job = Job.getInstance();
job.setJarByClass(AverageDriver.class);
job.setJobName("Max temperature");
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new
Path(args[1]));
job.setMapperClass(AverageMapper.class);
job.setReducerClass(AverageReducer.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);
System.exit(job.waitForCompletion(true) ? 0 : 1); }
}

AverageMapper package temp; import java.io.IOException; import
org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.LongWritable; import
org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Mapper; public class
AverageMapper extends Mapper<LongWritable, Text, Text, IntWritable> { public
static final int MISSING = 9999; public void map(LongWritable key, Text value,
Context context) throws IOException,
InterruptedException { int
temperature;
String line = value.toString();
if (line.length() < 93) {
return;

```
        }
        String year = line.substring(15, 19); if (line.charAt(87)
        == '+') { temperature =
        Integer.parseInt(line.substring(88, 92));
        } else { temperature = Integer.parseInt(line.substring(87,
          92));
        }
        String quality = line.substring(92, 93); if (temperature !=
        MISSING && quality.matches("[01459]")) { context.write(new
        Text(year), new IntWritable(temperature));
        }
    }
}
```

AverageReducer package temp; import java.io.IOException; import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Reducer; public class AverageReducer extends Reducer<Text, IntWritable, Text, IntWritable> { public void reduce(Text key, Iterable<IntWritable> values,

```
                Context context) throws IOException, InterruptedException {
        int sum = 0; int count = 0; for
        (IntWritable value : values) {
        sum += value.get(); count++;
        } if (count > 0) { int average = sum / count;
        context.write(key, new IntWritable(average));
        }
    }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>hadoop jar C:\avgtemp.jar temp.AverageDriver /input_dir/temp.txt /avgtemp_outputdir
2021-05-15 14:52:50,635 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-15 14:52:51,005 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2021-05-15 14:52:51,111 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1621060230696_0005
2021-05-15 14:52:51,735 INFO input.FileInputFormat: Total input files to process : 1
2021-05-15 14:52:52,751 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-15 14:52:53,073 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1621060230696_0005
2021-05-15 14:52:53,073 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-15 14:52:53,237 INFO conf.Configuration: resource-types.xml not found
2021-05-15 14:52:53,238 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-15 14:52:53,312 INFO impl.YarnClientImpl: Submitted application application_1621060230696_0005
2021-05-15 14:52:53,352 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1621060230696_0005/
2021-05-15 14:52:53,353 INFO mapreduce.Job: Running job: job_1621060230696_0005
2021-05-15 14:53:06,640 INFO mapreduce.Job: Job job_1621060230696_0005 running in uber mode : false
2021-05-15 14:53:06,643 INFO mapreduce.Job:  map 0% reduce 0%
2021-05-15 14:53:12,758 INFO mapreduce.Job:  map 100% reduce 0%
2021-05-15 14:53:19,860 INFO mapreduce.Job:  map 100% reduce 100%
2021-05-15 14:53:25,967 INFO mapreduce.Job: Job job_1621060230696_0005 completed successfully
2021-05-15 14:53:26,096 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=72210
                FILE: Number of bytes written=674341
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=894860
                HDFS: Number of bytes written=8
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
        Job Counters
                Launched map tasks=1
                Launched reduce tasks=1
                Data-local map tasks=1
                Total time spent by all maps in occupied slots (ms)=3782
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -ls /avgtemp_outputdir
Found 2 items
-rw-r--r--   1 Anusree supergroup          0 2021-05-15 14:53 /avgtemp_outputdir/_SUCCESS
-rw-r--r--   1 Anusree supergroup          8 2021-05-15 14:53 /avgtemp_outputdir/part-r-00000

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /avgtemp_outputdir/part-r-00000
1901    46

C:\hadoop-3.3.0\sbin>
```

b) find the mean max temperature for every month

MeanMaxDriver.class package
meanmax; import
org.apache.hadoop.fs.Path; import
org.apache.hadoop.io.IntWritable; import
org.apache.hadoop.io.Text; import
org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class MeanMaxDriver {

```
  public static void main(String[] args) throws Exception {
    if (args.length != 2) {
      System.err.println("Please enter the input and output parameters"); System.exit(-1);
    }
    Job job = Job.getInstance();
    job.setJarByClass(MeanMaxDriver.class);
    job.setJobName("Mean and Max Temperature");
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new
    Path(args[1]));
    job.setMapperClass(MeanMaxMapper.class);
    job.setReducerClass(MeanMaxReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    System.exit(job.waitForCompletion(true) ? 0 : 1); }
}
```

MeanMaxMapper.class package meanmax; import java.io.IOException; import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.LongWritable; import org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Mapper; public class MeanMaxMapper extends Mapper<LongWritable, Text, Text, IntWritable> { public static final int MISSING = 9999; public void map(LongWritable key, Text value, Context context) throws IOException, InterruptedException {

```
    int temperature;
    String line = value.toString(); if
    (line.length() < 93) return;
    String month = line.substring(19, 21); if (line.charAt(87) == '+') {
      temperature = Integer.parseInt(line.substring(88, 92));
    } else { temperature = Integer.parseInt(line.substring(87,
      92));
    }
    String quality = line.substring(92, 93); if (temperature !=
    MISSING && quality.matches("[01459]")) { context.write(new
    Text(month), new IntWritable(temperature));
    }
  }
}
```

MeanMaxReducer.class package meanmax; import java.io.IOException; import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Reducer; public class MeanMaxReducer extends Reducer<Text, IntWritable, Text, IntWritable> {

```
  public void reduce(Text key, Iterable<IntWritable> values, Context context) throws
      IOException, InterruptedException {
```

```
        int maxTemp = 0; int totalTemp
        = 0; int count = 0; int days = 0;
        for (IntWritable value : values) {
        int temp = value.get(); if (temp >
        maxTemp) {
              maxTemp = temp;
           } count++; if (count == 3)
           { totalTemp += maxTemp;
           maxTemp = 0; count = 0;
           days++;
           } } if (days
        > 0) {
           context.write(key, new IntWritable(totalTemp / days));
        }     else     {     context.write(key,     new
        IntWritable(0));
        }
    }
}
```

OUTPUT

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /meanmax_output/*
01      4
02      0
03      7
04      44
05      100
06      168
07      219
08      198
09      141
10      100
11      19
12      3

C:\hadoop-3.3.0\sbin>
```

LAB 7

For a given Text file, Create a Map Reduce program to sort the content in an alphabetic order listing only top 10 maximum occurrences of words

TopNDriver.class

```
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.util.GenericOptionsParser;
public class TopN {
public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    String[] otherArgs = (new GenericOptionsParser(conf, args)).getRemainingArgs();
    if (otherArgs.length != 2) {
      System.err.println("Usage: TopN <in> <out>");
      System.exit(2);
    }
    Job job = Job.getInstance(conf);
    job.setJobName("Top N");
    job.setJarByClass(TopN.class);
    job.setMapperClass(TopNMapper.class);
    job.setReducerClass(TopNReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(otherArgs[0]));
    FileOutputFormat.setOutputPath(job, new Path(otherArgs[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1); }
}
```

TopNCombiner.class

```
package samples.topn;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
```

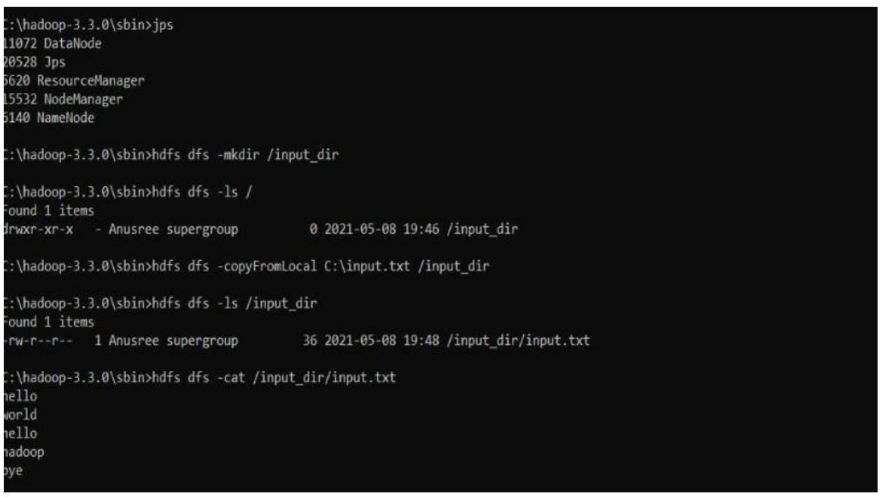
public class TopNCombiner extends Reducer<Text, IntWritable, Text, IntWritable> { public void reduce(Text key, Iterable<IntWritable> values, Context context) throws IOException, InterruptedException {
int sum = 0; for (IntWritable val : values) sum += val.get();
context.write(key, new IntWritable(sum)); }
}

TopNReducer.class package samples.topn; import java.io.IOException; import java.util.HashMap; import java.util.Map; import java.util.LinkedHashMap; import java.util.stream.Collectors; import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Reducer; public class TopNReducer extends Reducer<Text, IntWritable, Text, IntWritable> { private Map<Text, Integer> countMap = new HashMap<>(); public void reduce(Text key, Iterable<IntWritable> values, Context context) throws
IOException, InterruptedException {
int sum = 0; for (IntWritable val : values) { sum += val.get();
} countMap.put(new Text(key), sum);
}
protected void cleanup(Context context) throws IOException, InterruptedException {
Map<Text, Integer> sortedMap = countMap.entrySet().stream().sorted(Map.Entry.<Text, Integer>comparingByValue().reversed()).limit(20).collect(Collectors.toMap(
Map.Entry::getKey,
Map.Entry::getValue,
(e1, e2) -> e1,
LinkedHashMap::new)); for (Map.Entry<Text, Integer> entry : sortedMap.entrySet()) { context.write(entry.getKey(), new IntWritable(entry.getValue()));
}
}
}

TopNMapper.class package samples.topn; import java.io.IOException; import java.util.StringTokenizer; import org.apache.hadoop.io.IntWritable; import org.apache.hadoop.io.Text; import org.apache.hadoop.mapreduce.Mapper; public class TopNMapper extends Mapper<Object, Text, Text, IntWritable> { private static final IntWritable one

```
= new IntWritable(1); private Text word = new Text(); private String tokens =
"[_|$#<>\\^=\\[\\]\\*/\\\\,;,.\\-:()?!\"']";
        public void map(Object key, Text value, Mapper<Object, Text, Text, IntWritable>.Context
context) throws IOException, InterruptedException
            {
        String cleanLine = value.toString().toLowerCase().replaceAll(this.tokens, " ");
        StringTokenizer itr = new StringTokenizer(cleanLine); while
        (itr.hasMoreTokens()) { this.word.set(itr.nextToken().trim());
        context.write(this.word, one);
        }
    }
}
```

OUTPUT



```
C:\hadoop-3.3.0\sbin>jps
11072 DataNode
20528 Jps
6620 ResourceManager
15532 NodeManager
6140 NameNode

C:\hadoop-3.3.0\sbin>hdfs dfs -mkdir /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - Anusree supergroup          0 2021-05-08 19:46 /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -copyFromLocal C:\input.txt /input_dir

C:\hadoop-3.3.0\sbin>hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r--   1 Anusree supergroup         36 2021-05-08 19:48 /input_dir/input.txt

C:\hadoop-3.3.0\sbin>hdfs dfs -cat /input_dir/input.txt
hello
world
hello
hadoop
bye
```

```
C:\hadoop-3.3.0\sbin>hadoop jar C:\sort.jar samples.topn.TopN /input_dir/input.txt /output_dir
2021-05-08 19:54:54,582 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2021-05-08 19:54:55,291 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Anusree/.staging/job_1620483374279_0001
2021-05-08 19:54:55,821 INFO input.FileInputFormat: Total input files to process : 1
2021-05-08 19:54:56,281 INFO mapreduce.JobSubmitter: number of splits:1
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1620483374279_0001
2021-05-08 19:54:56,552 INFO mapreduce.JobSubmitter: Executing with tokens: []
2021-05-08 19:54:56,843 INFO conf.Configuration: resource-types.xml not found
2021-05-08 19:54:56,843 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2021-05-08 19:54:57,387 INFO impl.YarnClientImpl: Submitted application application_1620483374279_0001
2021-05-08 19:54:57,507 INFO mapreduce.Job: The url to track the job: http://LAPTOP-JG329ESD:8088/proxy/application_1620483374279_0001/
2021-05-08 19:54:57,508 INFO mapreduce.Job: Running job: job_1620483374279_0001
2021-05-08 19:55:13,792 INFO mapreduce.Job: Job job_1620483374279_0001 running in uber mode : false
2021-05-08 19:55:13,794 INFO mapreduce.Job:  map 0% reduce 0%
2021-05-08 19:55:20,020 INFO mapreduce.Job:  map 100% reduce 0%
2021-05-08 19:55:27,116 INFO mapreduce.Job:  map 100% reduce 100%
2021-05-08 19:55:33,199 INFO mapreduce.Job: Job job_1620483374279_0001 completed successfully
2021-05-08 19:55:33,334 INFO mapreduce.Job: Counters: 54
        File System Counters
                FILE: Number of bytes read=65
                FILE: Number of bytes written=530397
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=142
                HDFS: Number of bytes written=31
                HDFS: Number of read operations=8
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=2
                HDFS: Number of bytes read erasure-coded=0
```

```
C:\hadoop-3.3.0\sbin>hdfs dfs -cat /output_dir/*
hello   2
hadoop  1
world   1
bye     1

C:\hadoop-3.3.0\sbin>
```

LAB 8&9

1. Write a Scala program to print numbers from 1 to 100 using for loop.

```
object ExampleForLoop1 { def
   main(args: Array[String]) {
   for(counter <- 1 to 100) {
   print(counter + " ")
      }
      println()
   }
}
```

OUTPUT

2. Using RDD and FlatMap count how many times each word appears in a file and write out a list of words whose count is strictly greater than 4 using Spark.

```
from pyspark import SparkContext sc =
SparkContext("local", "SimpleWordCount")

# Read file rdd = sc.textFile("file.txt") # Replace with
your filename

# Count words counts = (rdd.flatMap(lambda
line: line.split())
          .map(lambda word: (word, 1))
          .reduceByKey(lambda a, b: a + b)
          .filter(lambda x: x[1] > 4))

# Show result for word, count in
counts.collect():
   print(word, count)

sc.stop()
```

OUTPUT

```
scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[9] at textFile at <console>:24

scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[11] at textFile at <console>:24

scala> val mapdata = splitdata.map(word => (word, 1))
mapdata: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[12] at map at <console>:25

scala> val reducedata = mapdata.reduceByKey(_ + _)
reducedata: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[13] at reduceByKey at <console>:25

scala> val filteredWords = reducedata.filter{ case (word, count) => count > 4 }
filteredWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[14] at filter at <console>:25

scala> filteredWords.collect.foreach(println)

scala> val data = sc.textFile("/home/bmscecse/Desktop/sparkdata.txt")
data: org.apache.spark.rdd.RDD[String] = /home/bmscecse/Desktop/sparkdata.txt MapPartitionsRDD[16] at textFile at <console>:24

scala> val splitdata = data.flatMap(line => line.split(" "))
splitdata: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[17] at flatMap at <console>:25

scala> val mapdata = splitdata.map(word => (word, 1))
mapdata: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[18] at map at <console>:25

scala> val reducedata = mapdata.reduceByKey(_ + _)
reducedata: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[19] at reduceByKey at <console>:25

scala> val filteredWords = reducedata.filter{ case (word, count) => count >= 4 }
filteredWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[20] at filter at <console>:25

scala> filteredWords.collect.foreach(println)
(hello,4)

scala> filteredWords.saveAsTextFile("/home/bmscecse/Desktop/filtered_words")

scala> val sortedWords = reducedata.sortBy(_._2, ascending = false)
sortedWords: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[26] at sortBy at <console>:25

scala> sortedWords.collect.foreach(println)
(hello,4)
(spark,3)
(world,2)
(big,1)
(data,1)
(,1)

scala> val countWords = filteredWords.count()
countWords: Long = 1

scala> println(s"Number of words with count >= 4: $countWords")
Number of words with count >= 4: 1

scala> 
```

LAB 10

Write a simple streaming program in Spark to receive text data streams on a particular port, perform basic text cleaning (like white space removal, stop words removal, lemmatization, etc.), and print the cleaned text on the screen. (Open Ended Question).

```
import org.apache.spark.SparkConf import
org.apache.spark.streaming.{Seconds, StreamingContext}
import org.apache.spark.streaming.StreamingContext._

object SimpleStreamingCleaner {
  def main(args: Array[String]): Unit = {
    // Create Spark configuration and StreamingContext val conf = new
    SparkConf().setMaster("local[2]").setAppName("StreamingTextCleaner") val ssc = new
    StreamingContext(conf, Seconds(5))

    // Connect to socket stream (e.g., using: `nc -lk 9999`) val
    lines = ssc.socketTextStream("localhost", 9999)

    // Define stop words val stopWords = Set("a", "an", "the", "is", "in", "on", "and", "of",
    "to", "this", "that")

    // Define basic lemmatization map (for demo only) val lemmaMap = Map("running" ->
"run", "tests" -> "test", "files" -> "file", "words" -> "word")

    // Clean the text stream val cleaned =
    lines.map(_.toLowerCase.trim) // lower case and trim
      .flatMap(_.split("\\s+"))  // split into words
      .filter(word => !stopWords.contains(word)) // remove stop words
      .map(word => lemmaMap.getOrElse(word, word)) // basic lemmatization

    // Print the cleaned text cleaned.print()

    // Start the stream ssc.start()
    ssc.awaitTermination()
  }
}
```

OUTPUT

```
scala> import java.nio.file.{Files, Paths}
import java.nio.file.{Files, Paths}

scala>

scala> val path = "/home/bmscecse/Desktop/WC.txt"
path: String = /home/bmscecse/Desktop/WC.txt

scala> val exists = Files.exists(Paths.get(path))
exists: Boolean = true

scala> println(s"File exists? $exists")
File exists? true

scala> val textFile = sc.textFile("file:///home/bmscecse/Desktop/WC.txt")
textFile: org.apache.spark.rdd.RDD[String] = file:///home/bmscecse/Desktop/WC.txt MapPartitionsRDD[40] at textFile at <console>:26

scala>

scala> val words = textFile.flatMap(line => line.split("\\s+"))
words: org.apache.spark.rdd.RDD[String] = MapPartitionsRDD[41] at flatMap at <console>:27

scala>

scala> val pairs = words.map(word => (word, 1))
pairs: org.apache.spark.rdd.RDD[(String, Int)] = MapPartitionsRDD[42] at map at <console>:27

scala>

scala> val counts = pairs.reduceByKey(_ + _)
counts: org.apache.spark.rdd.RDD[(String, Int)] = ShuffledRDD[43] at reduceByKey at <console>:27

scala>

scala> counts.collect().foreach{ case (word, count) => println(s"$word: $count") }
this: 3
is: 2
some: 1
: 1
hello: 5
testing: 1
just: 1
file: 3
test: 2
world: 4
contains: 1
a: 1
for: 1
words: 1

scala>
```